

Flight Radar – Operation

The device shows aircraft in the surrounding area. The round display is the radar; the small display shows details of the selected aircraft. Power on: connect the 5 V power supply. The radar appears after about 30 seconds.

Controls (rotary encoder)

Action	Effect
Turn (display shows ZOOM)	Change range: 10, 25, 50, 100 or 200 km
Short press	Next page on the small display (1/3, 2/3, 3/3, then back to 1/3)
Turn (display shows SELECT)	Select an aircraft. The selected aircraft is marked red and its data appears on the small display.
Long press (about 1 second)	Switch between ZOOM and SELECT (shown at the bottom of the radar)
No input for 30 seconds (in SELECT)	Returns to ZOOM automatically; the nearest aircraft is selected

Procedure: selecting an aircraft

1. Press and hold the encoder – the bottom of the radar shows **SELECT**.
 2. Turn the encoder until the desired aircraft is red.
 3. If needed, press the encoder briefly to see more data.
- The selection stays on the same aircraft between data updates.

Display

Radar (round display)	Meaning
Blue triangle	Own location (center)
Green dot	Aircraft
Red dot with ring	Selected aircraft
Rings / number at top	Distance; the number applies to the outer ring
N, E, S, W	North, east, south, west

Small display	Content
Page 1/3	Callsign, route (ICAO airports), altitude (ft), speed (kt), heading, distance
Page 2/3	Operator, manufacturer and type, climb/descent rate (ft/min), ICAO address
Page 3/3	IP address, aircraft within range and within 200 km, nearest aircraft, age of data

Setup (first start, new WiFi or location)

1. On first start both displays show **SETUP**. To open setup later, hold the encoder pressed while powering on.
 2. Connect a phone or PC to the WiFi **FlightRadar-Setup** (no password). Open **192.168.4.1** in the browser.
 3. Select the WiFi network, enter the password and save. Only the WiFi is required. Enter latitude/longitude and the OpenSky credentials (client ID, secret) later at **flightradar.local**.
 4. Tap **Save**. The device restarts and shows the radar.
- In the home network the page is also available at **flightradar.local** (or at the IP address on page 3/3).

Notes

- Data updates every 30 seconds, so the dots move in steps. “No traffic” means no aircraft within the radius: select a larger range.
- If the device does not respond: unplug the power supply, wait 10 seconds, plug it in again. 1 ft = 0.30 m, 1 kt = 1.85 km/h. WiFi is required.

ESP32 Pinout

ESP32 DevKit (38 pin, AZ-Delivery). Run all components at 3.3 V. Disconnect the ESP32 from power while wiring.

Component	Component pin	ESP32 pin	Done
Round display GC9A01 (SPI)	VCC	3V3	
	GND	GND	
	SCL (SCK)	GPIO18	
	SDA (MOSI)	GPIO23	
	CS	GPIO15	
	DC	GPIO2	
	RES	GPIO4	
	BLK	3V3	
OLED SSD1306 (I2C, 0x3C)	VCC	3V3	
	GND	GND	
	SDA	GPIO21	
	SCL	GPIO22	
Rotary encoder EC11	+	3V3	
	GND	GND	
	CLK	GPIO25	
	DT	GPIO26	
	SW	GPIO27	
Power supply 5 V DC	+5 V	5V	
	GND	GND	

Check by GPIO

GPIO	2	4	15	18	21	22	23	25	26	27
Function	DC	RES	CS	SCK	SDA OLED	SCL OLED	MOSI	CLK	DT	SW

Used: GPIO 2, 4, 15, 18, 21, 22, 23, 25, 26, 27. The round display and the OLED share no pins.

Firmware update (OTA)

1. Set an **OTA password** (at least 6 characters) on the setup page. Without a password, updates are disabled.
2. Arduino IDE: select the network port **flightradar** under **Tools > Port**, upload and enter the OTA password.
3. Or open **flightradar.local/update** in a browser (user **admin**, password = OTA password) and upload the **.bin** file (**Sketch > Export Compiled Binary**).
4. The small display shows the progress, then the device restarts.

The first OTA-capable firmware must be flashed over USB. If the sketch does not fit the update partition, select **Tools > Partition Scheme > Minimal SPIFFS (1.9MB APP with OTA)**.

Notes

- Power: regulated 5 V supply (at least 1 A) to the 5V and GND pins of the ESP32. Disconnect it while flashing over USB.
- Underside: space for adhesive anti-slip pads, 10 mm diameter.
- GPIO2 (DC) and GPIO15 (CS) are strapping pins. If the upload fails, briefly disconnect both wires.
- BLK is tied to 3V3 (GPIO21 is used for the OLED SDA).
- OLED stays dark and the monitor prints "SSD1306 init failed": check SDA/SCL, change 0x3C to 0x3D in the sketch.
- If one detent of the encoder skips two steps: in `handleEncoder()` change the threshold 4 to 2.
- Settings are entered on the setup page; nothing to edit in the sketch. Serial Monitor: 115200 baud.